How BloodLink works, and how to test all of it

A blood-donation platform built around registered **blood banks**. Three roles, one spine: a **member** asks for blood, the system ranks compatible members by availability and distance, a hospital **officer** matches one or more of them until the request is fulfilled, everyone is kept informed by **notifications**, and anyone can find the shortest route to the nearest bank. Sections 01–08 explain the system; Section 09 lists the test data, and Section 10 gives a step-by-step script for **every flow** in the app.

WHAT'S NEW IN V7

A working **notification bell** for every role (unread badge, live refresh, mark as read) · the **Contact** form now saves messages to an admin **Messages** inbox · registration ends on a clear welcome card · the database now matches the app: **11 tables**, with the old student-logbook tables and the stray `donor_app.sql` dump removed · a flow-by-flow test script for all 51 flows.

---

01 / The cast

Three roles, at a glance

Everyone signs in through one of two doors — the admin portal (`login.php`) or the shared user portal (`user-login.php`, role dropdown: Member or Hospital Officer). The session role gates every page; opening another role's page sends you back to your own dashboard.

SESSION ROLE: ADMIN

Administrator

Runs the bank network: blood banks, members, hospital officers and per-bank stock; posts emergency alerts; oversees requests; reads reports; answers **contact messages**.

SESSION ROLE: OFFICER

Hospital Officer

Belongs to one blood bank. Works the request queue and the Emergency Board, matches requests to the best-scored available members, **confirms each donation**, and records **walk-in donations** that add units to their bank's stock.

SESSION ROLE: MEMBER

Member

One account that both **donates** and **requests**. Registers, sets availability, checks the 56-day rule, files requests, gets notified about emergencies and their requests, keeps a profile, and routes to the nearest bank.

WHY A SINGLE "MEMBER" ACCOUNT

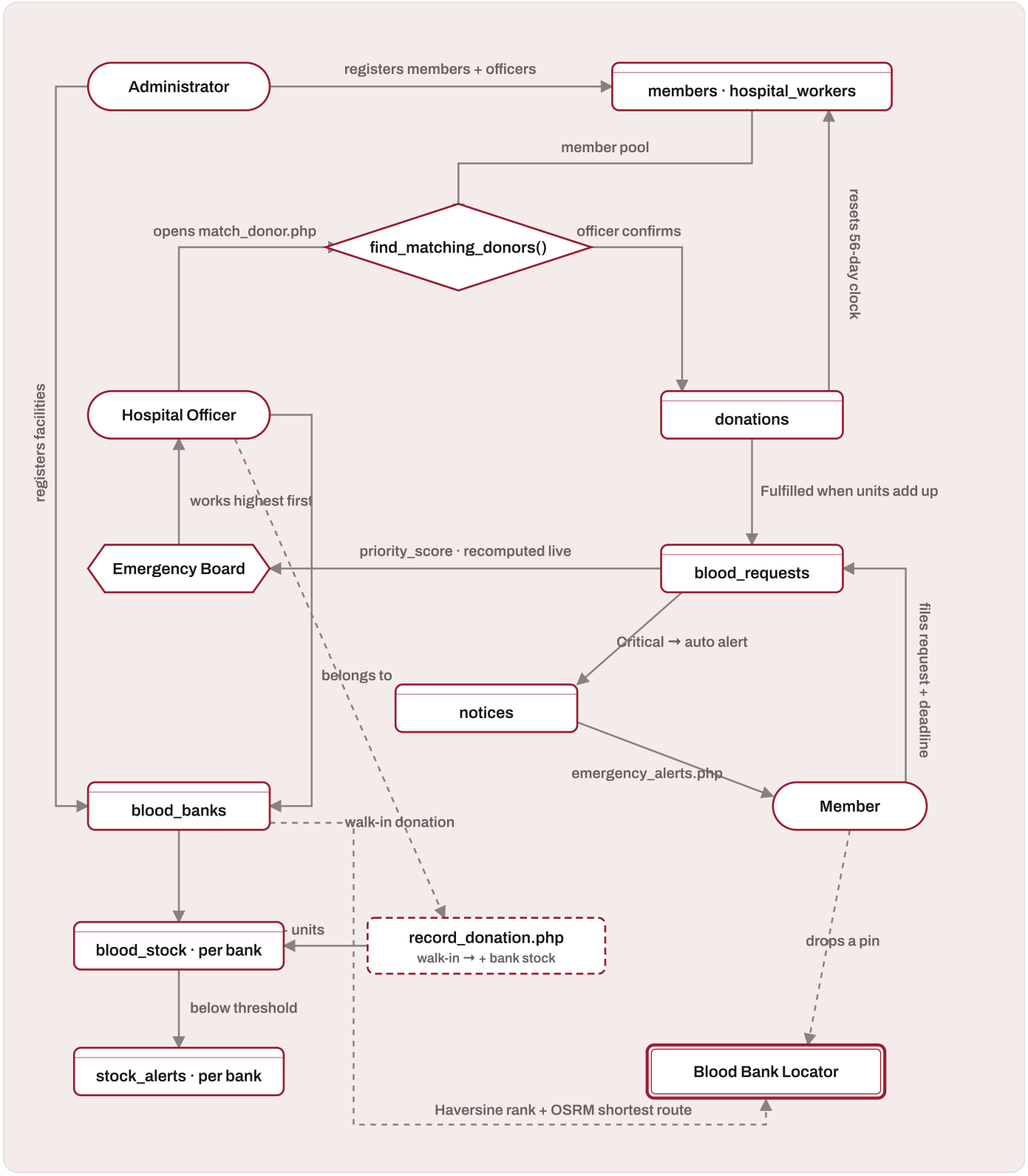
A real person can both give and need blood, so BloodLink uses **one member account for both** — every member is a searchable donor *and* can file requests. On top of that: a **donor availability registry**, an **emergency workflow** ranked by a live priority score, **compatibility-aware matching**, **analytics**, the **Blood Bank Locator** (Section 06) and **notifications** (Section 07).

02 / The spine

How the whole app connects

Everything orbits one workflow: a request becomes one or more matched, recorded donations. The locator, the 56-day rule, availability, per-bank stock and notifications all hang off it.

- **Admins** register the `blood_banks`, the people, and each bank's `blood_stock`.
- **Members** register once and create `blood_requests` with an urgency and optional deadline. A Critical Emergency also posts a `notices` alert and notifies every member who can donate.
- **Officers** belong to a bank. The Emergency Board ranks open requests by a live priority score. `match_donor.php` runs `find_matching_donors()`; `process_donation()` records each donation in one transaction and marks the request Approved or — once the units add up — Fulfilled.
- **Walk-in donations** (`record_donation.php`) add units to the officer's bank stock.
- Each of these events writes rows to `notifications` for the people involved (Section 07).


Administrator
registers members + officers
members · hospital_workers
member pool
resets 56-day clock
opens match_donor.php
find_matching_donors()
officer confirms
registers facilities
Hospital Officer
donations
works highest first
Fulfilled when units add up
Emergency Board
priority_score · recomputed live
blood_requests
Critical → auto alert
files request + deadline
belongs to
notices
emergency_alerts.php
Member
blood_banks
walk-in donation
units
record_donation.php
walk-in → + bank stock
drops a pin
blood_stock · per bank
below threshold
stock_alerts · per bank
Blood Bank Locator
Haversine rank + OSRM shortest route

03 / Data model

Eleven tables — every one in use

TABLE	HOLDS	WRITTEN BY	READ BY
`login`	Admin accounts	seed	admin login
`blood_banks`	Facilities (unique name, address, geocoded lat/lng, contact)	admin	locator, stock, officers, reports
`hospital_workers`	Officers (required `blood_bank_id`, position, unique email)	admin	officer login, donations, reports
`members`	Donor + recipient accounts: blood group, 56-day clock, lat/lng, availability, unique `member_code` and email	registration, admin, My Profile	login, matching, requests, reports
`blood_requests`	Requests: urgency, severity, deadline, live `priority_score`, status; `recipient_id` → the member who asked	members	queue, board, match, history
`donations`	Ledger — donor's blood group, units, bank; `request_id` NULL for walk-ins	`process_donation()`	My Donations, histories, reports
`blood_stock`	Units per bank per blood type	admin, walk-ins	stock page, dashboards, reports
`stock_alerts`	Low-stock alerts per bank per type	stock page, walk-ins	dashboards, stock page
`notices`	Emergency alerts (plain text)	officers, admins, critical requests	member Emergency Alerts
`notifications`	Per-user bell items: recipient role + id, type, title, body, link, read flag	events in Section 07	the bell, Notifications page
`contact_messages`	Messages from the public Contact page; new / read	Contact page	admin Messages inbox

REMOVED

`attendance`, `chat_messages`, `logbook_entries`, `projects`, `seminar`, `staff`, `students`, `weekly_summaries`, `allocation`, `timetable`, `donors` and `recipients` came from the earlier student-logbook version and the old donor/recipient split. Nothing in BloodLink uses them, so `schema.sql` and the upgrade script drop them, and the stale `db/donor_app.sql` dump and `uploads/*.csv` files are deleted. Members with donations and officers who confirmed donations can't be deleted, so the ledger is never broken.

04 / Role walkthroughs

What each role does, end to end

ADMINISTRATOR runs the network

Signs in at `login.php` → `dashboard.php`. Sidebar: Dashboard · Manage Members · Hospital Officers · Blood Banks · Blood Requests · Emergency Board · Find Donors · Blood Stock · Emergency Alerts · Geo-Map · Reports · **Messages**.

HOSPITAL OFFICER makes the match

Signs in at `user-login.php` → `officer_dashboard.php` (bank stats + stock). Sidebar: Dashboard · Donors · Blood Requests · Emergency Board · Find Donors · **Record Donation** · Emergency Alerts · Geo-Map.

MEMBER is both supply and demand

Registers at `register.php`, signs in with email or Member ID → `member_dashboard.php`. Sidebar: Dashboard · Donate Blood · Request Blood · Request History · **My Donations** · Emergency Alerts · Nearby Banks · **My Profile**.

Every role has the **notification bell** in the top bar, and the avatar menu (Dashboard · Notifications · My Profile for members · Logout). Section 10 walks through every one of these pages as a test.

05 / The enhancements

Four features that lift it above a basic register

01 · AVAILABILITY REGISTRY

Members say when they can give

Each member keeps an `is_available` flag, a schedule and a contact preference. **Find Donors** searches the pool by group and mode, optionally available-only, ranked from a chosen bank.

02 · EMERGENCY WORKFLOW

Requests are triaged, not just queued

A request carries a severity and an optional deadline. The **Emergency Board** recomputes each `priority_score` from the time left, and critical requests notify eligible donors directly.

03 · COMPATIBILITY-AWARE MATCHING

The system scores every donor — and enforces it

A compatibility score rewards exact matches and universal O−; available members sort first. On confirmation the server re-checks compatibility, eligibility and availability and records the **donor's** blood group.

04 · ANALYTICS

At-a-glance charts

Stock by blood group on the admin dashboard; Reports adds blood-group distribution, donation status, KPIs and CSV export; the home page shows live counts.

HOW THE PRIORITY SCORE IS COMPUTED

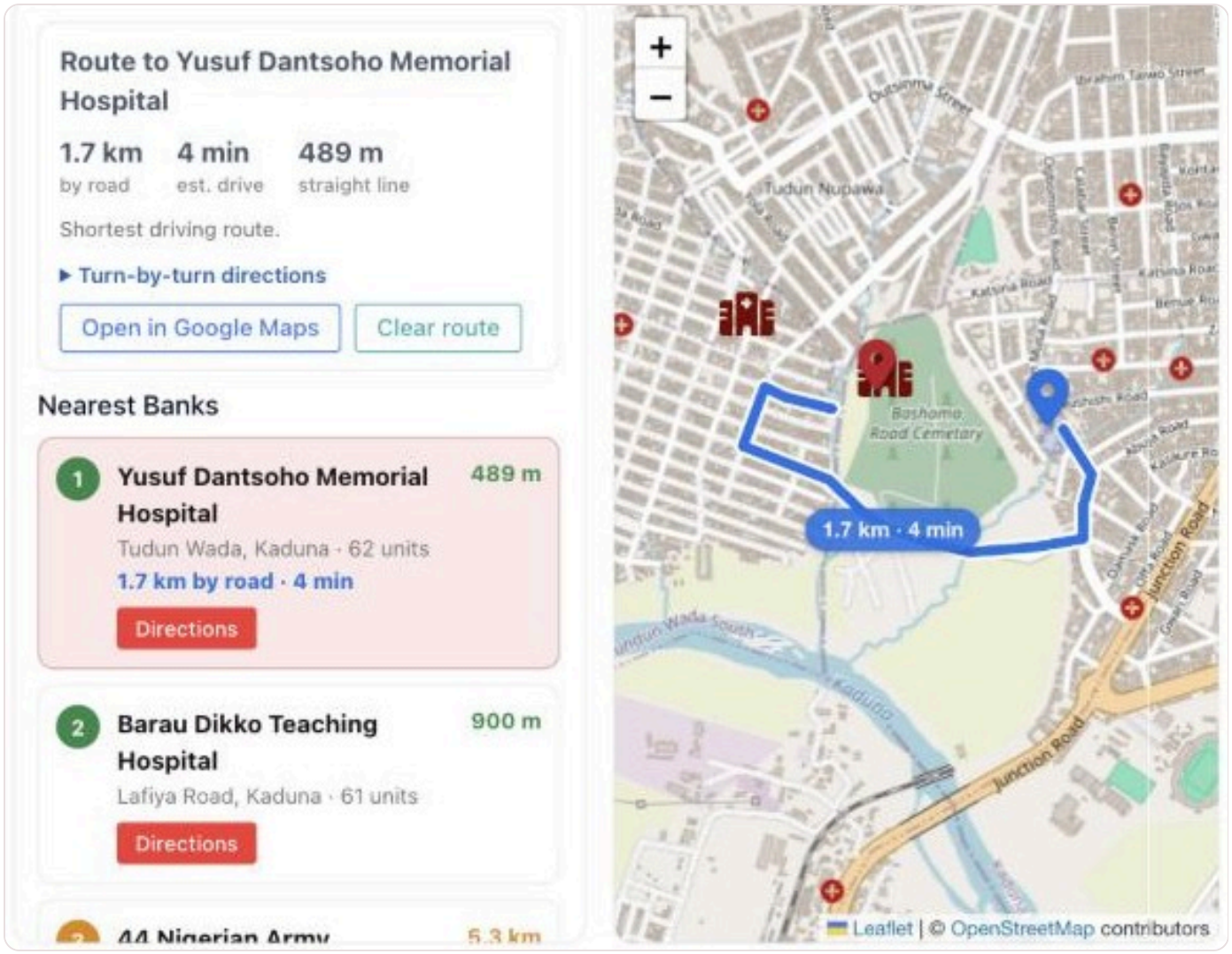
```
priority = severity       (critical 60 · severe 40 · moderate 20 · low 5)
         + rarity × 2     (AB− rarest 10 … O+ most common 3)
         + time pressure  (hours left ≤3: 20 · ≤12: 12 · ≤48: 6 · else 2)
         → capped at 100   # hours left = deadline − time elapsed, refreshed on every Board load
```

Compatibility score: 0 if incompatible, else 100 (exact) / 90 (donor O−) / 80 (same ABO) / 70 (otherwise compatible).

06 / Blood Bank Locator

The interactive map (brief Fig 5.5)

Answers "which registered bank is nearest, and what is the shortest way there?" for every role.



Blue pin = you; red pin = the chosen bank. The shortest road route is blue, labelled with kilometres and drive time; alternatives are grey dashes.

1. **Drop your pin**

   Use my location (GPS, with an accuracy ring), **Use Kaduna city centre**, or click anywhere on the map. Drag the pin to adjust.

2. **Banks are ranked**

   Straight-line distance (Haversine) sorts the list nearest-first with a coloured badge (green ≤ 5 km, amber 5–15, red > 15).

3. **The shortest route is drawn**

   OSRM returns up to three road routes; the shortest by distance is drawn in blue and labelled on the line. Click a grey route to use it instead.

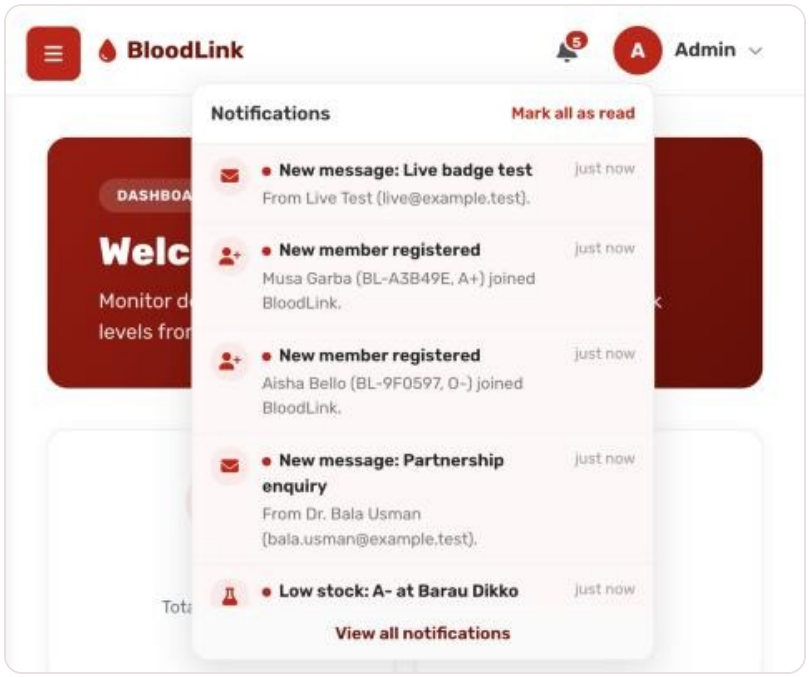
4. **Follow it**

   The panel shows road km, drive time, straight-line km, turn-by-turn directions and an "Open in Google Maps" link. If routing is down, a straight line is shown.

07 / Notifications & messages

Who hears about what

The bell in the top bar shows a red badge with your unread count and the latest eight notifications; it refreshes every 30 seconds. Clicking one opens the related page and marks it read; **Mark all as read** clears the badge; **View all notifications** lists everything.



The admin's bell with five unread notifications.

WHEN THIS HAPPENS	WHO IS NOTIFIED	CLICKING OPENS
`member registers`	Admins ("New member registered"); the new member ("Welcome to BloodLink")	Edit Member · Dashboard
`request filed`	The requester ("Request received"), every officer, admins	Request History · Match Donor · Emergency Board
`critical request`	Also every compatible, eligible, available member except the requester	Emergency Alerts
`donation confirmed`	The donor ("Thank you..."); the requester ("Your blood request is Approved / Fulfilled")	My Donations · Request History
`walk-in recorded`	The donor	My Donations
`stock below threshold`	Officers at that bank and admins — once per alert	Officer Dashboard · Blood Stock
`alert posted`	Every member	Emergency Alerts
`contact message`	Admins ("New message: ...")	Messages

Contact messages are stored in `contact_messages` and read in the admin **Messages** inbox, which shows new messages first and offers **Reply by email**, **Mark as read** and **Delete**.

`08 / The end-to-end donation`

Follow one unit of blood

1. **A member files a Critical O− request**
   → a Pending `blood_requests` row with a high priority, an automatic alert, and notifications to officers, admins and every eligible O− member.
2. **Officer opens the Emergency Board**
   the request is at the top, its countdown running → Match Donor (or straight from the notification).
3. **System proposes donors**
   only O−, 56-day-eligible members, available first, the requester excluded.
4. **Officer confirms (1–2 units)**
   `process_donation()` re-checks everything, records the donation with the donor's group and the officer's bank, resets the donor's clock, and marks the request Approved or Fulfilled.
5. **Everyone is told**
   the donor gets a thank-you; the requester sees the new status in their bell and Request History; reports update.

09 / Test setup & data

Set up once, then run Section 10

FRESH DATABASE

```
mysql -u root -proot -e "CREATE DATABASE IF NOT EXISTS donor_app"
mysql -u root -proot donor_app < db/schema.sql
mysql -u root -proot donor_app < db/seed.sql      # re-run both any time to reset the test data
```

EXISTING DATABASE (V5 OR EARLIER)

```
mysql -u root -proot donor_app < db/upgrade_2026_10_qa.sql   # safe to run more than once
```

In Local: open the site → **Database** tab → **Open Adminer** → select `donor_app` → **Import** → choose the file → **Execute**.

ACCOUNTS

ROLE	PAGE	SIGN IN WITH	PASSWORD
Admin	`login.php`	`admin`	`admin`
Officer	`user-login.php`	`maryam@bloodlink.test` — Barau Dikko (bank 1)	`Officer@123`
Officer	`user-login.php`	`samuel@bloodlink.test` — ABU Zaria (bank 4)	`Officer@123`
Members	`user-login.php`	any email below, or its Member ID	`Member@123`

SEEDED MEMBERS — PICK THE RIGHT ONE FOR EACH TEST

MEMBER	EMAIL · ID	GROUP	USE IT TO TEST
Zainab Auwal	zainab@ · BL-KD-001	O−	never donated, available — universal donor
Kamalu Yahaya	kamalu@ · BL-KD-002	O+	eligible donor; walk-in donations
Aliyu Abubakar	aliyu@ · BL-KD-003	A+	donated 30 days ago → **not eligible**
Fatima Sani	fatima@ · BL-KD-004	A−	donated exactly 56 days ago → eligible
Ibrahim Musa	ibrahim@ · BL-KD-005	B+	55 days ago and **unavailable**
Hauwa Garba	hauwa@ · BL-KD-006	B−	eligible; partial donations (O5)
Yusuf Danladi	yusuf@ · BL-KN-007	AB+	incompatible with B+ requests
Amina Bello	amina@ · BL-KN-008	AB−	**unavailable**
Grace Okonkwo	grace@ · BL-KD-009	O+	no address (not on the map); files M4
Hassan Umar	hassan@ · BL-KD-010	O−	has requests #1 (Critical O−) and #4 (Fulfilled)
Blessing Eze	blessing@ · BL-ZR-011	AB+	owns request #2 (AB+, 1 unit)
Musa Lawal	musa@ · BL-KN-012	B+	owns request #3 (B+, 3 units, Approved)

All emails end in `@bloodlink.test`. Barau Dikko (bank 1) starts with A− and O− below threshold. The seed also adds a few unread notifications and one contact message, so the bell and inbox have content on first login.

10 / Flow-by-flow test scripts

Every flow, step by step

Reset the data first (Section 09). Each flow says who runs it, where to start, what to do and what you should see. Flows marked with an officer or admin action can be run in a second browser window, signed in as that role.

P1	Browse the public site
P2	Send a contact message
P3	Register as a member
P4	Registration is validated
P5	Log in as a member
P6	Log in as an officer
P7	Log in as the administrator
P8	Log out
P9	Pages are protected
M1	Set donor availability
M2	Check donation eligibility
M3	File a normal blood request
M4	File a critical emergency
M5	Request validation
M6	Track requests
M7	See your donation history
M8	Read emergency alerts
M9	Edit your profile
M10	Change your password
M11	Find the nearest blood bank

M12	Use notifications
01	Read your bank dashboard
02	Work the Emergency Board
03	Open a request to match
04	Confirm a donation
05	Fulfil a request with several donors
06	Unsafe donations are refused
07	Record a walk-in donation
08	Search the donor pool
09	Browse the donor directory
010	Post and delete an emergency alert
011	Officer notifications
A1	Read the admin dashboard
A2	Add a member
A3	Edit a member
A4	Delete a member
A5	Add a hospital officer
A6	Delete an officer
A7	Manage blood banks
A8	Manage blood stock
A9	Oversee requests
A10	Post or remove any alert
A11	Reports and CSV export
A12	Export members
A13	Find donors and use the locator
A14	Read contact messages
A15	Admin notifications
X1	Forms reject forged submissions
X2	Phone layout
X3	Every page loads
X4	Upgrade an existing database

Public & sign-in P1–P9

P1 Browse the public site

WHO Visitor (logged out) START `index.php`

DO

1. Open the home page.
2. Click **About**, then **Contact**, then **Home**.
3. Click **Get Started**.

EXPECT

- ✓ Home shows live figures: **Registered Members**, **Units Donated**, **Registered Blood Banks** (seed: 12 · 0 · 5).
- ✓ About describes three portals: Administrators, Hospital Officers and Members.
- ✓ **Get Started** opens the registration form.

P2 Send a contact message

WHO Visitor START `contact.php`

DO

1. Fill **Your Name**, **Your Email**, **Subject** and a message of 10+ characters.
2. Click **Send Message**.
3. Now submit with a bad email and a 5-character message.

EXPECT

- ✓ A valid message shows "Thank you — your message has been sent" and the form is empty.
- ✓ Admins get a **New message** notification; it appears in the admin **Messages** inbox (A14).
- ✓ Invalid input lists each problem and keeps what was typed; nothing is saved.

P3 Register as a member

WHO Visitor START `register.php`

TEST DATA any new email, e.g. `aisha@bloodlink.test`, password `Donate2026`, address `Barnawa, Kaduna`.

DO

1. Enter name, email (mixed case is fine), phone, blood group, address, and a password with letters + numbers, typed twice. Leave **Member ID** blank.
2. Click **Submit Registration**.
3. Click **Log in now** and sign in with the Member ID shown.

EXPECT

- ✓ A "Welcome to BloodLink" card shows the new Member ID (e.g. `BL-3F9A21`); the form is gone and refreshing doesn't re-submit.
- ✓ The email is stored in lowercase and the phone as digits only; a known address is located on the map.
- ✓ The new member lands on their dashboard with 1 unread notification ("Welcome to BloodLink"). Admins get "New member registered".

P4 Registration is validated

WHO Visitor START `register.php`

TEST DATA existing email `zainab@bloodlink.test`.

DO

1. Submit once for each case: an email already registered (try different capitals), Member ID `BL-KD-001`, blank name, phone `12`, password `abc`, mismatched passwords, Member ID containing `@`.

EXPECT

- ✓ Each is refused with a specific message; nothing is saved.
- ✓ Typed values (except passwords) stay in the form.
- ✓ An unknown address still registers, with a note to fix it later in My Profile.

P5 Log in as a member

WHO Member START `user-login.php`

TEST DATA `zainab@bloodlink.test` or `bl-kd-001` / `Member@123`.

DO

1. Choose **Member (Donor / Recipient)**.
2. Log in with an email; log out; log in again with the Member ID in lowercase.
3. Try a wrong password.

EXPECT

- ✓ Both logins open `member_dashboard.php` immediately.
- ✓ A wrong password shows "Invalid email/member ID or password!" and keeps the role selected and the login typed.

P6 Log in as an officer

WHO Officer START `user-login.php`

TEST DATA `maryam@bloodlink.test` (Barau Dikko) or `samuel@bloodlink.test` (ABU Zaria) / `Officer@123`.

DO

1. Choose **Hospital Officer**, enter email and password, click **Login**.
2. Try submitting without choosing a role.

EXPECT

- ✓ Opens `officer_dashboard.php` showing the officer's blood bank.
- ✓ No role chosen → "Choose whether you are a Member or a Hospital Officer."

P7 Log in as the administrator

WHO Admin START `login.php` (or the round icon on the home page)

DO

1. Enter `admin` / `admin`, click **Login**.

EXPECT

- ✓ Opens `dashboard.php` immediately. A wrong password shows "Invalid Login Credentials!"

P8 Log out

WHO Any role START any signed-in page

DO

1. Click the avatar (top right) → **Logout**; repeat with **Logout** in the sidebar.

EXPECT

- ✓ Returns to the home page; pressing Back and refreshing a dashboard sends you to a login page.

P9 Pages are protected

WHO Visitor, then each role START type URLs directly

DO

1. Logged out, open `dashboard.php`, `officer_dashboard.php`, `member_dashboard.php`.
2. As a member, open `reports.php`; as an officer, open `manage_members.php`.
3. While logged in, open `login.php` or `register.php`.

EXPECT

- ✓ Admin pages send logged-out visitors to `login.php`; other pages to `user-login.php`.
- ✓ A wrong role is sent back to its own dashboard — no data is shown.
- ✓ Logged-in users are sent to their dashboard instead of the login/register pages.

Member M1–M12

M1 Set donor availability

WHO Member START Dashboard → **My Donor Availability**

DO

1. Untick **Available to donate**, set **Schedule** to Weekends and **Contact preference** to SMS.
2. Click **Update Availability**, then refresh the page.

EXPECT

- ✓ "Availability updated." shows once; refreshing doesn't resubmit; the values are kept.
- ✓ An officer matching a request now sees this member with a disabled **Unavailable** button (O4).

M2 Check donation eligibility

WHO Member START Sidebar → **Donate Blood**

TEST DATA `zainab@…` (never) and `aliyu@bloodlink.test` (30 days ago).

DO

1. Open the page as a member who has never donated, then as one who donated 30 days ago.

EXPECT

- ✓ Never donated → "You are currently eligible to donate".
- ✓ 30 days ago → not eligible, with the days remaining (26).

M3 File a normal blood request

WHO Member START Sidebar → **Request Blood**

DO

1. Choose **Normal**; fill patient name, blood group, units, hospital, location and phone.
2. Click **Submit Request**, then refresh the page.

EXPECT

- ✓ "Your blood request has been submitted successfully"; refreshing doesn't submit it again.
- ✓ The request shows as Pending in Request History (M6).
- ✓ The member gets "Request received"; every officer gets "New … request" linking to Match Donor; admins are notified too.

M4 File a critical emergency

WHO Member START **Request Blood**

TEST DATA file as `grace@bloodlink.test` (O+); `zainab@…` and `hassan@…` (O−) should be notified.

DO

1. Choose **Critical Emergency**, set **Needed within** 2 hours, blood group O−, 1 unit, hospital Barau Dikko Teaching Hospital, Kaduna.
2. Submit.

EXPECT

- ✓ An emergency alert is posted automatically (Emergency Alerts, M8).
- ✓ Every compatible, eligible, available member (O− here) gets an "Emergency: O− needed..." notification — not the requester, and not ineligible or incompatible members.
- ✓ The request sits at or near the top of the Emergency Board (O2).

M5 Request validation

WHO Member START **Request Blood**

DO

1. Submit with no patient name and 0 units.
2. Submit a second Pending request for the same blood group.

EXPECT

- ✓ Each problem is listed ("Patient name is required.", "Units needed must be at least 1.").
- ✓ "You already have a pending ... request" — the duplicate is refused.

M6 Track requests

WHO Member START Sidebar → **Request History**

TEST DATA `hassan@bloodlink.test` has seeded requests.

DO

1. Open the page after an officer confirms a donation for your request (O4).

EXPECT

- ✓ Each request shows urgency and status (Pending → Approved → Fulfilled); **Donations Received** lists the units and donor.

M7 See your donation history

WHO Member START Sidebar → **My Donations**

DO

1. Open the page after an officer records a matched (O4) or walk-in (O7) donation for you.

EXPECT

- ✓ Tiles show donations recorded, units given, and when you can donate again.
- ✓ Each row shows date, group, units, who it was for (or "Walk-in donation") and the blood bank.

M8 Read emergency alerts

WHO Member START Sidebar → **Emergency Alerts**

DO

1. Open after an officer posts an alert (O10) or a critical request is filed (M4).

EXPECT

✓ Alerts list newest first with who posted them; text such as "Ben & Jerry" shows exactly as typed.

M9 Edit your profile

WHO Member START Sidebar → **My Profile**

DO

1. Change phone and address; click **Save Details**.
2. Try changing the email to one another member uses.

EXPECT

✓ "Profile saved."; the address is re-located (it says "Located on the map").

✓ A taken email is refused. Blood group is shown but not editable here.

M10 Change your password

WHO Member START **My Profile** → Change password

DO

1. Enter a wrong current password; then the right one with a new password typed twice.
2. Log out and back in with the new password.

EXPECT

✓ Wrong current password is refused; a correct change shows "Password changed." and the new password works.

M11 Find the nearest blood bank

WHO Member (also officers and admins) START Sidebar → **Nearby Banks** / **Geo-Map**

DO

1. Click **Use Kaduna city centre** (or **Use my location**).
2. Click somewhere else on the map; then drag the blue pin.
3. Click a different bank in the list; then click a grey dashed route if one is shown.
4. Open **Turn-by-turn directions** and **Open in Google Maps**.

EXPECT

✓ A blue pin marks you; banks re-rank nearest-first; the shortest road route to the nearest bank is drawn in blue with "x km · y min" on the line.

✓ Moving the pin re-routes to the new nearest bank — unless you picked a bank yourself, which stays selected.

✓ A grey route switches the panel to "Alternative route — n km longer". Directions list each turn; Google Maps opens with the same start and end.

M12 Use notifications

WHO Any role START the bell in the top bar

DO

1. Click the bell.
2. Click one notification.
3. Open the bell again → **Mark all as read**; then **View all notifications**.
4. Leave a page open while someone triggers a notification for you (e.g. an officer posts an alert).

EXPECT

- ✓ The badge shows the unread count; unread items have a red dot.
- ✓ Clicking opens the related page and marks it read (the badge drops by one).
- ✓ Mark all clears the badge; View all lists every notification.
- ✓ Within about 30 seconds the badge and list update without reloading.

Hospital officer 01–011

01 Read your bank dashboard

WHO Officer START `officer_dashboard.php`

TEST DATA Maryam (Barau Dikko) sees A− and O− highlighted as low.

DO

1. Log in and look at the top of the dashboard.

EXPECT

- ✓ Shows your bank name, open requests, critical emergencies, low-stock groups at your bank, units you recorded this month, and your bank's stock by group (low groups highlighted).

02 Work the Emergency Board

WHO Officer (and admin) START Sidebar → **Emergency Board**

DO

1. Open the board; wait a minute.

EXPECT

- ✓ Open requests are ranked by priority (severity + rarity + time left), each with a ticking countdown; the most urgent is first.
- ✓ Priorities are recomputed on every visit, so a request climbs as its deadline nears.

03 Open a request to match

WHO Officer START Sidebar → **Blood Requests**

DO

1. Click **Match Donor** on a request.

EXPECT

- ✓ The match page shows the request, units still needed, and only compatible, 56-day-eligible members ranked available-first, by match score, then distance.
- ✓ The person who filed the request is never listed.

04 Confirm a donation

WHO Officer START `match_donor.php`

TEST DATA donor `Zainab Auwal` (O−).

DO

1. On request #2 (AB+, 1 unit), click **Confirm Match** next to an O− donor and confirm.

EXPECT

- ✓ "Donation recorded (1 unit(s) of O−). Request is now Fulfilled." — recorded under the **donor's** group.
- ✓ The donor gets "Thank you — your donation was recorded"; the requester gets "Your blood request is Fulfilled".
- ✓ The request leaves the queue; opening it again shows "no further donations can be recorded". Bank stock is unchanged (directed blood goes to the patient).

05 Fulfil a request with several donors

WHO Officer START `match_donor.php?request_id=3` (B+, 3 units)

TEST DATA B− donor `Hauwa Garba`, then O+ donor `Grace Okonkwo`.

DO

1. Confirm 2 units from one donor.
2. Confirm the last unit from another donor.

EXPECT

- ✓ After the first: status Approved and "1 still needed"; the units box is capped at 1.
- ✓ After the second: Fulfilled. The requester is told each time.

06 Unsafe donations are refused

WHO Officer START `match_donor.php`

TEST DATA unavailable: `Amina Bello`, `Ibrahim Musa`.

DO

1. Look for **Confirm Match** next to an unavailable member.
2. Try 3 units in one donation.
3. Re-confirm a donor who just gave.

EXPECT

- ✓ Unavailable members show a disabled **Unavailable** button.
- ✓ Units are limited to 1–2 per donation and never above what's still needed.
- ✓ A donor inside the 56-day window is refused with the days remaining. (The server also refuses incompatible donors and a member donating to their own request.)

07 Record a walk-in donation

WHO Officer START Sidebar → **Record Donation**

TEST DATA `kamalu@bloodlink.test` (O+).

DO

1. Enter a member's email or Member ID and 1 unit; click **Record Donation** and confirm.
2. Try the same member again; try an unknown email.

EXPECT

- ✓ "Walk-in donation recorded: 1 unit(s) of … added to your bank's stock." and it appears under Recent walk-in donations.
- ✓ Your bank's stock for that group rises (O1); a low-stock alert clears once stock is back above threshold.
- ✓ The member is thanked and sees it in My Donations. A repeat is refused by the 56-day rule; unknown members show "No member found".

08 Search the donor pool

WHO Officer (and admin) START Sidebar → **Find Donors**

DO

1. Choose O−, **Compatible donors**, a bank under **Distance from**, tick **Available only**, click **Search**.
2. Switch to **Exact group only**; untick Available only.

EXPECT

- ✓ Results show group, match score, eligibility, availability and contact, ranked eligible → best match → nearest.
- ✓ Exact mode lists only that group; unticking includes unavailable members.

09 Browse the donor directory

WHO Officer (and admin) START Sidebar → **Donors**

DO

1. Open the list.

EXPECT

✓ Every member with Member ID, blood group and 56-day eligibility.

010 Post and delete an emergency alert

WHO Officer START Sidebar → **Emergency Alerts**

DO

1. Enter a title and message; click **Post Emergency Alert**.
2. Click **Delete** on it and confirm.

EXPECT

✓ "Emergency alert posted successfully!"; every member gets an "Emergency alert: ..." notification and sees it on their Emergency Alerts page.

✓ Delete removes it. Officers can delete only their own alerts; admins can delete any.

011 Officer notifications

WHO Officer START the bell

DO

1. Have a member file a request (M3), then have the admin drop a stock level at your bank below threshold (A8).

EXPECT

✓ "New ... request" links straight to that request's Match Donor page.

✓ "Low stock: ... at your bank" — only officers at that bank receive it.

Administrator A1–A15

A1 Read the admin dashboard

WHO Admin START `dashboard.php`

DO

1. Log in.

EXPECT

✓ Totals for members, officers, pending requests and active low-stock alerts, plus the stock-by-group chart.

A2 Add a member

WHO Admin START Sidebar → **Manage Members**

DO

1. Fill the form (leave Member ID blank) and click **Add Member**.
2. Try an existing email or Member ID.

EXPECT

- ✓ "Member added (ID BL-...)."; the new member can log in (P5).
- ✓ Duplicates and invalid fields are refused with the same rules as registration.

A3 Edit a member

WHO Admin START Manage Members → **Edit**

DO

1. Change the phone, untick **Available to donate**, set a **New Password**; click **Update Member**.

EXPECT

- ✓ "Member updated."; the member now shows as unavailable when matching, and can log in with the new password. Leaving New Password blank keeps the old one.

A4 Delete a member

WHO Admin START Manage Members → **Delete**

DO

1. Delete a member with no donations; then try one who has donated.

EXPECT

- ✓ The first is removed. The second is kept with a message explaining why (deleting would erase the donation ledger) — untick Available instead.

A5 Add a hospital officer

WHO Admin START Sidebar → **Hospital Officers** → **Add Hospital Officer**

DO

1. Fill name, email, phone, position, choose a blood bank, set a password; save.
2. Try without a bank, and with an existing officer email.

EXPECT

- ✓ "Hospital Officer ... added."; they can log in (P6) and their dashboard shows that bank.
- ✓ Missing bank and duplicate email are refused.

A6 Delete an officer

WHO Admin START Hospital Officers → **Delete**

DO

1. Delete a new officer; then try one who has confirmed donations.

EXPECT

- ✓ The first is removed. The second is kept with "can't be deleted without breaking the donation ledger".

A7 Manage blood banks

WHO Admin START Sidebar → **Blood Banks**

DO

1. Add a bank with a Kaduna address; edit it; delete it.
2. Try adding a name that already exists.

EXPECT

- ✓ A new bank is located on the map (Locator) and gets 8 empty stock rows.
- ✓ Duplicate names show "A blood bank with that name already exists."

A8 Manage blood stock

WHO Admin START Sidebar → **Blood Stock**

DO

1. Pick a bank; set A+ to 1 with threshold 5; click **Update**.
2. Click Update again; then set A+ to 12.
3. Try −3 units.

EXPECT

- ✓ "Stock updated successfully."; a low-stock alert appears and that bank's officers + admins are notified — once, not on every save.
- ✓ Restocking resolves the alert. Negative numbers are refused.

A9 Oversee requests

WHO Admin START Sidebar → **Blood Requests** and **Emergency Board**

DO

1. Open both pages.

EXPECT

- ✓ The full open queue and the ranked board are visible; confirming donations is left to officers by design.

A10 Post or remove any alert

WHO Admin START Sidebar → **Emergency Alerts**

DO

1. Post an alert; delete one an officer posted.

EXPECT

- ✓ Admin posts show as "BloodLink Admin"; admins can delete any alert. Members are notified of new alerts.

A11 Reports and CSV export

WHO Admin START Sidebar → **Reports**

DO

1. Set **From**/**To** dates and **Apply Filter**.
2. Click **Export CSV Report** and open the file.

EXPECT

- ✓ KPIs and charts (blood-group distribution, donation status) reflect the date range.
- ✓ The CSV opens as a clean spreadsheet — a header row and one row per donation, no page HTML.

A12 Export members

WHO Admin START Manage Members → **Export to Excel**

DO

1. Click the export button and open the file.

EXPECT

✓ `members_list.xls` opens with every member, group, availability and last donation.

A13 Find donors and use the locator

WHO Admin START **Find Donors**, **Geo-Map**

DO

1. Repeat O8 and M11 as admin.

EXPECT

✓ Same results as for officers and members.

A14 Read contact messages

WHO Admin START Sidebar → **Messages**

TEST DATA the seed includes one message from "Dr. Bala Usman".

DO

1. Open the inbox after P2.
2. Click **Mark as read** on one, **Reply by email** on another, **Delete** on a third.

EXPECT

✓ New messages show first with a NEW tag; the sidebar shows the new-message count.

✓ Mark as read removes the tag; Reply opens your mail app addressed to the sender with "Re: subject"; Delete removes the message.

A15 Admin notifications

WHO Admin START the bell

DO

1. After P2, P3, M3 and A8, open the bell.

EXPECT

✓ "New message", "New member registered", "New … request" and "Low stock" notifications, each opening the right page.

Cross-cutting X1–X4

X1 Forms reject forged submissions

WHO Tester START any form

DO

1. In the browser dev tools, change the hidden `csrf_token` value on a form and submit.

EXPECT

✓ "Invalid or expired form token" (HTTP 419); nothing changes.

X2 Phone layout

WHO Any role START any page at phone width (≈375 px)

DO

1. Shrink the browser or use a phone.

EXPECT

✓ The sidebar becomes a menu button; the logo, bell and avatar don't overlap; the map sits above the bank list; nothing scrolls sideways.

X3 Every page loads

WHO Tester START each sidebar link, every role

DO

1. Click every sidebar link for admin, officer and member with the browser console open.

EXPECT

✓ Every page loads with no PHP warnings and no console errors.

X4 Upgrade an existing database

WHO Tester START a database from v5 or earlier

DO

1. Run `db/upgrade_2026_10_qa.sql` (Section 09 setup), then run it a second time.

EXPECT

✓ Both runs finish without errors; afterwards the database has exactly the 11 tables in Section 03 and the old logbook tables are gone.

KNOWN LIMITATIONS (EXPECTED)

- **Nominatim** allows about one lookup per second: saving many addresses quickly may leave some without map coordinates until re-saved.
- The public **OSRM** demo server suits a demo, not production traffic; self-host it for real deployment.
- Browsers share GPS location only on **HTTPS** or `localhost`; elsewhere use the Kaduna button or click the map.
- Notifications are in-app only (no email or SMS); "Reply by email" opens the admin's own mail app.
- Admins don't confirm donations by design — the officer is the clinical actor, and the donation's bank is the officer's bank.